

CHAPTER 6

TRAINING PLANS



CLIMBING PERFORMANCE IS dependent on many factors. If we can place these factors in a training plan, we will see more progression than if we only do the same things every time we train. By planning our training, we can train both specifically and with variety to achieve our goals, and at the same time manage the training load so that we avoid overtraining and thereby decrease our risk of injury.

In this chapter, we look at basic training principles, training periodisation, and how we make both long-term and short-term training plans.

Planning for training is based on a few fundamental principles:

THE PRINCIPLE OF LOAD AND ADAPTATION

The body is an amazing organism that will over time adapt to any load we expose it to. This is how muscles grow and become stronger through strength training, and how we can recover better during rest phases through endurance training. This is also how moves become steadily easier through technique training, and how experience helps us deal with stressful competition situations. This happens by training – and challenging ourselves on – the elements we wish to improve. Be prepared for the adaptation to take time!

Through well-planned and varied training it is also easier to manage the training load, something which is central to reducing the risk of injury. Training load is a combination of volume, intensity and frequency – how much, how hard and how often you train. If you climb hard and often, your training load will be significant. If, on the other hand, you can reduce the volume of the more intensive sessions and the intensity of the volume sessions, it can be easier to maintain a level of training load that allows you to train a lot without overloading your body.

Different types of tissues in your body also require different amounts of time to adapt. Muscles adapt quickly, while bones, tendons and ligaments need longer to adapt to the same load. Hard and uniform training with too rapid a progression can quickly lead to the tendons being loaded more than they are trained to endure, which we know can lead to injury. Having a proper time perspective on the relationship between load and adaptation is therefore central to planning for training.

THE PRINCIPLE OF PROGRESSION

Despite the body being great at adaptation, it is by nature lazy. When we don't challenge ourselves more than what we can already endure or achieve, there is no need for the body to adapt further. Therefore, progression in training is a very important element when setting up a training plan. We create progression by gradually increasing the load

or exercising an ever-increasing range of movements.

WITHOUT PROGRESSION, THERE'S ONLY STAGNATION.

THE PRINCIPLE OF RECOVERY

After training, rest is required so that the body is able to perform again. Most people who have had two consecutive days of long and hard bouldering or endurance sessions will have experienced that the second day was harder than the first. Although the recovery time will differ depending on how well trained you are initially, and what and how hard you trained, it's a general rule of thumb that after a session of hard strength or endurance training you should wait about two days before you repeat a similar session. This does not mean that you should rest completely for two days, but that you can train other things in the meantime. Recovery time depends on, among other things, general physical shape, sleep and nutrition. This explains why elite athletes in good physical shape, and who focus on sleeping and eating properly, can endure periods with a high number of extremely hard workouts. For us mortals, it is important to get enough rest between hard workouts and gradually build the foundation needed to endure more and harder workouts. Poor recovery results in poorer quality of exercise, or as the Finnish scientist Keijo Häkkinen says:

If you go too hard on your easy days, soon you will be going too easy on your hard days.

THE PRINCIPLE OF SPECIFICITY

It is a well-known fact that you become better at whatever you are training. If you want to get better at climbing long, pumpy routes, you need to simulate this in your training. If you want to become stronger to keep tension on steep climbs, you must train specifically for this – including using the exercises presented in this book, as more general strength training for the abdomen and back will not produce the same effect. As you develop as a climber, it is important to identify which areas you want to improve so that you can choose specific exercises and training methods.

THE PRINCIPLE OF VARIATION

In contrast to the principle of specificity, variation in your training is also important. For example, you should let your fingers rest the day after a finger-intensive bouldering session; instead, train endurance on better holds on day two. This way you spread the load throughout the body. Variation between types of moves, types of holds, wall angles, and indoor and outdoor climbing is also essential to building the technical skills necessary to become a better climber. In addition, new elements can increase your eagerness to train and progress further. By varying your training, you can also maintain the level of skills other than the ones you are focusing on for that period. You can, for example, focus on hard bouldering and strength for a period, but at the same time vary your training week by adding an endurance session and a more technical bouldering session, so that you maintain your level in these areas. This way you also get to train and climb more times that week, which undoubtedly will help make you a better climber.



PERIODISATION

Periodisation is a term that refers to planned variations in training to achieve specific goals, and periodisation of training is based on the training principles. We need to build a training plan that promotes progress through loading, adaptation and recovery, and with enough progression in the training to avoid stagnation; we also need to train specifically to improve on the various elements, and vary the training sufficiently to keep us motivated and include as many elements as possible.

For the physical factors – strength and endurance – it is common to start the training period with a greater focus on volume than on intensity. In strength training, this means a higher number of repetitions with a lower load per set, where the aim is to learn the exercise and build a base for harder repetitions. Eventually, the number of repetitions per set is reduced, but is replaced with a significantly heavier load per repetition. Thereafter, strength training can be focused more at rapid development of force.

The same approach can be used with endurance training. At the start of the training period, it makes sense to have a high number of moves per climbing session. So that you can complete all of these moves, they cannot be too hard, nor should you get too pumped. The goal is to build the capacity to later withstand more intensive endurance training. As the number of moves per session gradually decreases, the difficulty level per move, circuit or route can be increased, simultaneously increasing the intensity. The final part of an endurance period is about training as close to your intended performance level as possible. For a competitive climber, this means climbing a few, but very hard, routes per session, with maximum effort. If you have an outdoor project, the routes or circuits you train on can be similar to the route you are training for in the number of moves, cruxes and rests. You will also climb at maximum effort per attempt and have longer rests between each attempt.



PHOTO: ALEX MANELIS

Maria Davies Sandbu bouldering in Bishop, USA.

Planning your training in this way is called linear periodisation. Linear indicates that there is a progression from high volume and low intensity to low volume and high intensity. The challenge with such a model is that it goes on for a relatively long time, and you will prioritise some elements in your training over others. Combining a linear model with what we call conjugate periodisation allows you to incorporate training of other factors simultaneously. A conjugate model involves switching between different properties for shorter periods, and this can still be combined with the principle of progression from high volume and low intensity to low volume and high intensity. In an endurance period, around 80 per cent of your training can be built around endurance training, while the remaining 20 per cent may be aimed at bouldering and strength. In this way, it's possible to maintain strength during an endurance period and vice versa.

During each training period, it is appropriate to vary the training load over the different weeks. For example, two weeks of hard training may be followed by one easier week before another two hard weeks. This is known as load rhythm and it creates a positive relationship between load and recovery. How many hard and easy weeks you plan for during the period depends on your fitness level and what timescale you have set for when you want to be in top shape. If you have many hard weeks in a row, you will find that your physical condition gets progressively worse because you are training a lot, and thus you must lower your performance ambitions during such a period. When you are conscious of this and are confident that you will peak during easier weeks – and confident that you are actually training well even though you may feel weaker – it will be easier to live with fluctuations in fitness. The physically easier weeks are also golden opportunities to focus more on technical, tactical and mental factors. Thus, easy weeks do not mean you should not climb, and we encourage you to do as much valuable climbing as possible in the easier weeks even if the physical load is reduced.

PEAKING

The point of training will always be to make you a better climber, performing at ever-higher levels. Still, there will be times when you want to be able to give it that extra effort. If you are going on a climbing trip or trying to redpoint a route that you have been working on, you will want to engage that last percentage that separates a top from a fall, giving you the feeling of being weightless and transforming the previously impossible crux into something that is now suddenly easy. Timing peak fitness may seem easy on paper, but it takes a lot of experimentation to find the model that suits you best.

All exercise breaks the body down. After periods of hard training, you can ease up on the training load for a little longer than you normally would, creating what is called super-compensation. This means that you are raising your level for a period after training at considerably less load for a while. How much you raise the level, and how long that period lasts, depends on how hard and long you have trained and how well you have recovered.

The hardest thing about peaking is getting the timing right. In endurance sports, a peak period usually starts three weeks before reaching maximum performance, while in more strength-dependent sports, which includes climbing, the period starts two weeks in before. The most important thing is to reduce the training duration – how long each session lasts – and it is recommended to reduce the duration by 40 to 60 per cent. At the same time, the training frequency – how often you exercise – can remain unchanged, and you can train just as hard as before. Short and intense, i.e. hard, sessions are what appear to have the best effect on fitness development during such a period. The key is to load less, recover more and at the same time maintain your capacity.

If you exercise four times a week where each session is about 90 to 120 minutes, you can continue with the four sessions, but reduce their duration to 45 to 60 minutes per session at high intensity. Start with the reduction in training duration about two weeks before you plan to be at peak performance. To preserve the peak, continue with short, intense sessions – but at some point your fitness curve will start to drop and it's time to start a new training period. Therefore, it may be challenging to have multiple peaks in a short period of time, as each peak will require a pre-peak training period. You can use the same principles of load reduction if your goal is to get a small peak for the Sundays you plan to be out on your projects. Then you can do a hard workout on Monday and Tuesday, rest Wednesday and train two short, hard sessions on Thursday and Friday before resting on Saturday. The ideal method for peaking will vary between individuals, and we encourage everyone to experiment for themselves based on the recommendations above.



The last week before a bouldering trip or a competition, try doing short and hard bouldering and campus sessions every other day with two rest days before you want to start performing.

TRAINING PLANS

We often think of training plans as reserved for elite athletes who train a lot, and they do of course have a lot to gain from planning their training. But by considering the principles described above we can all get more out of our training if we have an idea of what to do, how to do it and why we should be doing it.

GOALS, REQUIREMENTS AND CAPACITY

All training plans are guided by the goals we set. We can set long-term dream goals, for example, a dream route or a competition result, or we can set more short-term, intermediate objectives along the way. Climbing a specific boulder problem, route or grade is an inspiring goal, but we recommend that you also set some broader goals. If your goal is to climb a certain F7c, you should at the same time have a goal to climb as many routes graded F7a to F7b on as many different wall angles as possible. Not only will you achieve your goal of climbing your dream route, but you will also have become a more solid climber at a higher level. It is important that your goals create both motivation for training and provide direction for what to train.

Once your goals have been set, the challenging task of planning the necessary training to achieve these goals can begin. For this task, you should consider your strengths and weaknesses – i.e. what you need to develop so you can achieve your goals. Some things will be easier to define than others. For example, it can be quite obvious to you if your fingers aren't strong enough to do a certain crux move, but it can be harder to define how technically correct you are moving or how efficiently you are clipping the quickdraws. Even when it comes to pure physical factors, it isn't always easy to define how good you must be when it comes to individual skills – i.e. how strong your fingers must be or how good your endurance must be – for you to achieve your goals. Forming a picture of where your strengths lie and where your greatest potential for development lies will therefore require effort. If you are a physically strong climber who thrives on steep terrain but you want to develop so you can climb routes and boulder problems that are more vertical and technically challenging, then these are the elements you need to prioritise in your training. Similarly, climbers who thrive on small holds and technically challenging climbing should prioritise strength training to develop as a climber. A simple exercise to start this process is to list and categorise what you see as your strengths and your potential for development in your climbing. It is a good idea to make the first draft of this list yourself, but we recommend that most climbers include their training partners or coaches in this process so it is as accurate as possible when defining where you are as a climber.

In the example opposite, we have split performance factors into technical, physical and mental, and made subcategories for each of them. As you go through each category, mark the ones you consider as strengths with green, and the ones where you need to put in more effort with red. You can use our example as a starting point, but challenge yourself to come up with and list each element on your own. This will strengthen your sense of ownership of your development plan and at the same time require you to reflect on your own strengths and potential for improvement.

TECHNICAL

Footwork	Edging	Smearing	Precision	Steep climbing	Heel hooking	Toe hooking
Wall angle	Slab	Vertical	Overhang	Roof		
Balance	Weight transfer	Flagging	Fronting	Twisting	Sidepull	
Climbing style	Static	Dynamic	Deadpoint	Dyno		
Tension	Compression	Not cutting loose on steep climbing				
Grip positions	Crimps	Slopers	Pockets	Pinches		

PHYSICAL

Finger strength	Finger-intensive bouldering	Fingerboard max		Contact strength
Arm and upper body strength	Pull-ups	Locking off	Campusing	Steep bouldering
Core	Overhangs/roofs	Swings	Not cutting loose on steep climbing	
Endurance	Power endurance	Maximum endurance	Capacity	Fingerboard endurance
Mobility	High foot placements	Splits/corners	Hips close to the wall when fronting	Stemming with the arms

MENTAL

Inner dialogue	Positive thinking	Confidence	Forgetting failure	Remembering success	
Motivated for	Outdoor climbing	Indoor climbing	Strength training	Technique training	Mental training
Visualisation	Redpoint	Reading routes/on-sight	Flash	New environments	
Performance	Outdoors	Indoors	Training		
Preparations	Routines		Time spent training		
Fear	Fear of falling	Fear of failing	Climbing in front of others		

When you have gone through each point you will have a picture of how you – and possibly others – see your strengths and weaknesses. It is, however, important to see the different properties in relation to one another. If, for example, you rate your maximum finger strength as good, yet still have problems with finger-intensive bouldering, it's probably not because your fingers aren't strong enough, but rather that you don't have good contact strength. In the same way, you can be strong when doing climbing-specific core exercises but still struggle with roof climbing. This is probably more to do with a lack of technique on that wall angle than purely a lack of strength.

Being aware of what you are already good at and what you want to develop adds to the quality of your training, where each session focuses on improving specific elements.



PHOTO: BJØRNAR SMESTAD



TINA JOHNSEN HAFSAAS:

- Spend time building a solid foundation.
- Do as much climbing as possible in each training session.
- Climb every style.
- Specific training only works if you have a good foundation.
- A solid foundation gives you the best conditions for getting stronger, developing better endurance and staying away from injuries.

Seven years ago, Tina Johnsen Hafsaas, Norwegian national champion and World Cup climber, completed this exercise. By becoming aware of what she was already good at, and where she had the biggest potential for growth, it was easier for Tina to add the different elements to her daily training routine. Analysing her own skillset became habit after a while, and this has made it possible for her to develop into an elite-level climber.

WEAKNESSES	STRENGTHS
Slopers	Crimps
Big pinches	Crimpy pinches
Pulling through on bad holds	Pulling through on good holds
Roof climbing	Vertical climbing
Chalking and resting too much	Finding rests
Heel hooking	Toe hooking
Locking off	Landing holds with bent elbows
Explosive moves	Twisting in – using the legs to relieve the arms
Fear of falling	Solving sequences on the fly
Shouldery moves	Reading and planning sequences
Burly moves	Dynamic moves (as long as I am in control)
Max power	Grit
Power endurance	Creating speed

Tina’s chart is more detailed than our examples and it also contains additional elements. And herein lies much of the value of this exercise: feel free to use our examples, but challenge yourself to find elements unique to your own climbing.

LONG-TERM PLANS

It makes sense to produce an overview of what will happen during the coming year. Depending upon your ambitions, long-term plans can also run for several years. For example, if you have been climbing for a few years but have the ambition to become a national champion, it's natural to consider your development over a longer period than just one year. For complete beginners, the most important thing will be to climb as much as possible without thinking too much about when to do what during the year. However, for more advanced climbers, it will be more important to define different performance periods and associated training periods which precede these.

In many countries, the year is traditionally split into an outdoor season and an indoor season, where the indoor season is characterised by indoor training while the outdoor season is all about trying – and hopefully sending – the routes and boulder problems we have dreamed about during the winter. As climbing destinations around the world have become more accessible, it is now more common to travel to places that allow climbing even in winter, and this has an effect on how we plan our training if we want to be fit to perform on these trips.

The fundamental message is that heavy training periods are rarely compatible with peak performance. You must, therefore, divide the year into different training and performance periods, and see how you can plan these around when you would like to be in top shape. If you do not have specific times of the year when you want to perform, and would rather focus on getting progressively better, it is still important to think about how different skills can be specifically developed and during which periods of the year this should happen.



PHOTO: ALEX MANELIS

Maria Davies Sandbu climbing in Hueco Tanks, USA, during winter.

PERIODIC PLANS

Based on your long-term plan you can define different periods throughout the year. The most important part of a periodic plan is to define the priority for that specific period. If one of your intermediate objectives is to become stronger in your upper body, then you must implement several sessions which target that; likewise if you wish to improve your endurance. At the same time, it is important to maintain the level of the elements that you aren't prioritising – for example, during periods of strength training you also maintain your endurance, and vice versa. A rule of thumb is to spend 80 per cent of your training time on what you wish to prioritise, and the remaining 20 per cent on the other elements. With every period you should plan for a gradual progression so that the training becomes progressively harder. It is also important to create a loading rhythm for every period so that you can alternate between harder and easier weeks and harder and easier sessions.

In the example below we have divided a 12-week period into three blocks of four weeks. Our example is for a climber who has enough time for three climbing days per week, and one day for strength training. The focus for the period is strength, so we have implemented strength training for the upper body, arms and fingers, in addition to regular climbing training. By organising the training in such a way, we have defined the types of sessions we want for each week and what they should contain. We have planned a loading rhythm per week and progressively harder training throughout the period. How the sessions are distributed throughout the week can vary based on multiple factors, so we recommend setting up the plan for each week to begin as close to the start of the week as possible.

WEEKS 1-4	SESSIONS PER WEEK	LOAD RHYTHM	FOCUS
	2 x Bouldering	Week 1: Medium	Focus for the weeks should be varied bouldering sessions, both technical, and finger intense and physical. The hard weeks involve a higher number of moves for the endurance sessions, and higher intensity for the bouldering sessions. The easy weeks involve a rest day between each session. Strength training focuses on a higher number of repetitions (8-12 x 3 sets).
	1 x Lead/circuits	Week 2: Hard	
	1 x Arm/upper body strength	Week 3: Hard	
		Week 4: Easy	
WEEKS 5-8	SESSIONS PER WEEK	LOAD RHYTHM	FOCUS
	2 x Bouldering	Week 1: Medium	More focus on intensive bouldering and endurance sessions. The hard weeks involve finger-intensive and physically challenging bouldering sessions. The easy weeks involve more technical bouldering and lower-intensity endurance sessions. Strength training focuses on fewer repetitions (4 x 4).
	1 x Deadhang	Week 2: Hard	
	1 x Lead/circuits	Week 3: Hard	
	1 x Arm/upper body strength	Week 4: Easy	
WEEKS 9-12	SESSIONS PER WEEK	LOAD RHYTHM	FOCUS
	2 x Bouldering	Week 1: Medium	Short and hard bouldering sessions with few but hard boulder problems and longer rests between each attempt. Endurance sessions with few attempts, but high intensity per route or circuit. Strength training with three sets of 2-3 repetitions at maximum intensity.
	1 x Deadhang	Week 2: Hard	
	1 x Lead/circuits	Week 3: Hard	
	1 x Arm/upper body strength	Week 4: Easy	

The first four weeks focus more on quantity, while the intensity gradually ramps up during the whole period. The plan alternates between hard and easy weeks, and every week involves both endurance training and more specific technique training in individual sessions. The period ends at high intensity and with more rest to culminate in a peak in fitness.

SHORT-TERM PLANS

When you have defined the focus of a training period it will be easier to decide the contents of your training week. You can plan the contents of a training period as we did in the examples opposite, but you always have to take illness, injuries and other commitments into account when planning the training week. It is, therefore, best to decide when the different sessions should take place during the week as close to the start of the training period as possible. You should also plan your training such that the most physically demanding sessions follow a rest day, in order to maximise the training dividend from these sessions. This means that campus training and other explosive sessions should come first, then the physically demanding strength sessions, and then lastly the endurance sessions before another rest day. Below are two weeks from the periodic plan so that we can take a more detailed look at each day of the week.

WEEK 3 HARD	MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY	SUNDAY
	Bouldering: 5–10 technical problems, top at least 5	Endurance: 3 x 5	Rest	Strength: pull-ups, lock-offs, hanging abdomen exercises	Rest	Bouldering: pyramids	Rest

WEEK 9 MEDIUM	MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY	SUNDAY
	Endurance: 2 x 4	Rest	Bouldering: Moonboard	Strength: pull-ups, lock-offs, hanging abdomen exercises	Rest	Deadhang max + 3–5 technical problems, top at least 3	Rest

When you are planning the sessions for a week, you need to know what each session should contain and how it should play out. The latter point is one of the most important things to learn about planning your training. The most common way is to divide the session into three: the warm-up, the main part and the finish.



PHOTO: LATTICE TRAINING

INTERVIEW WITH

TOM RANDALL

Tom Randall is one of the UK's top climbers and founder of Lattice Training which provides data-driven climbing assessments and training plans. A climber and coach for over 20 years, he made the first ascent of Century Crack (5.14b) in Utah, USA, and The Kraken (Font 8b) in Devon, England. He is also a former GB climbing team manager and coach. We asked Tom for his thoughts on his training philosophy and his approach to structured climbing training.

‘YEAR AFTER YEAR, I STILL GO BACK AND TRY TO MAKE IMPROVEMENTS ON BASIC THINGS LIKE STRENGTH, MENTAL GAME AND TECHNIQUE.’

1. You're known for training hard and it's paid off with some incredibly hard ascents around the world. What has your personal experience of climbing-specific training taught you over the years?

I think the biggest thing for me as a climber – but also as a coach – is that the process of being very specific and structured with my training will only take me so far with my performance. Using the element of specificity, or replica training, is the method by which you can finely tune your climbing to almost perfectly match up to the demands of your goal. It often feels like it's a bit of a performance hack because it's so incredibly effective, but it's really just a method of training that's been used by most sports for decades. It's exactly the type of work that you would do in other sports during what is called a 'peak cycle' or a 'performance taper', but it has only become really popular in climbing in the last ten or so years. That being said, it should be noted that it's like the polishing of a diamond and not a magic pill for success. If what you have at the fundamental core of your climbing ability is just a lump of coal, you're not going to get a precious stone, no matter how much you rub it! That's why, year after year, I still go back and try to make improvements on basic things like strength, mental game and technique. If I don't, I find myself stagnating and getting into a cycle of endless performance polishing, trying to squeeze out that last one per cent of perfection because I'm too lazy or ill-disciplined to tackle the underlying issues.

2. Training for climbing is becoming increasingly more structured and sophisticated. Are we now closing the gap with more established/mainstream sports?

Before climbing, my background was in athletics and martial arts and I won various national titles in both, so I've spent almost all of my life in elite sport. In addition, I stay connected with other sports coaches to try and understand what's going on in the more mainstream sports and sports science literature. When I combine what I've seen and experienced in both of these areas (as a junior athlete and as a coach), I'm left with the feeling that climbing isn't actually doing that bad at all! I think the climbing population is very self-critical when it comes to training and I see us excel in many areas compared to other sports. Yes, football may have access to a lot of money, cooler sports tech and coaching resources, but when I speak to the coaches about the fundamentals of muscle physiology and the understanding of movement economy and so on, I'm rarely left with the feeling that we're decades behind. One of the things we've worked very hard at with Lattice is taking the methods used by the best in other sports and deploying them in climbing. We've been mimicking the coaching structures and education, the training methods and research, and the data-backed approach for years now.



PHOTO: JON VICKERS

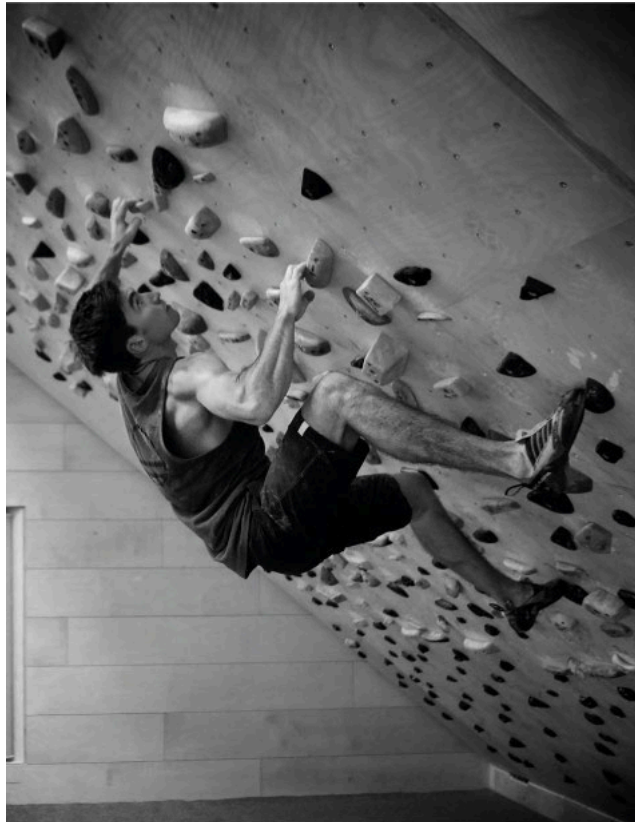


PHOTO: LATTICE TRAINING

‘I ALWAYS TRY TO REMIND CLIENTS THAT UNLESS THE TRAINING THEY’RE DOING HAS A HIGH LEVEL OF SPECIFICITY THEN THEY’RE CLOSE TO WASTING THEIR TIME.’

3. You founded Lattice in 2016. What is at the heart of the Lattice approach to climbing training?

Lattice is all about creating a coaching and performance ecosystem that will deliver results to teams or individuals whether they’re keen amateurs or the world’s elite. What I mean by this is that nothing at Lattice sits in isolation. We don’t want to have coaches or data or research that sits on its own trying to bring about the best performance. Everything is designed and run so that it fits together in this big organism that works off of all of its strengths in a cohesive way and that constantly aims to feed back into the core so that everyone and everything improves bit by bit over the years. For example, the training we do must be backed by sports science expertise, so we employ sports science graduates, master’s and PhDs. Then we ask them to educate and share their expertise and experience in an open and critical manner within the team. From there, we have resources and methodologies created for our clients, knowledge shared across multiple platforms of social media, and finally we’ll make sure we publish papers on what we’re doing for peer review. This is the approach that I’ve seen the elite coaching setups take elsewhere and myself and the whole team are totally driven to make this happen at Lattice.

4. What advice would you give someone who would like to start training seriously so that they can take their climbing to another level?

You’ve got two approaches really. You can self-educate and put a load of time into learning how to train (this book is a great place to start!) so that you can then go on your own self-coached journey. Alternatively, you can get yourself a coach who will normally be able to shortcut a lot of your time and – hopefully, if they know what they’re doing – make your progression more efficient on both time and injury risk. As a summary point on both of these approaches, it’s key to understand that the biggest factors that people fail to get right are training load, training specificity and consistency. When we talk about training load, we’re referencing the stress that we put our bodies under, and it is directly necessary to create adaptation through overload. I have seen many climbers over the years fail to grasp that load is a function of both intensity and volume and consequently they find themselves with poor results – or they quickly get injured. In reference to training specificity, I always try to remind clients that unless the training they’re doing has a high level of specificity then they’re close to wasting their time if it’s too big a portion of their weekly activity. Finally, consistency wins the day every time. You look at nearly all the best climbers and they’re the product of years of consistent progress and they typically don’t have much time out with injury. Those that do seem to be very good at creating methods of working around the problem.



WARMING UP

Warming up prepares you to perform physically, technically and mentally, and also reduces the risk of injury. The warm-up should consist of a general part and a specific part. The general part can be running, jumping, push-ups, squats and similar activities, where the goal is to activate the whole body and use the large muscle groups. The warm-up is also a good time to go through some injury prevention exercises where you train the movement patterns that are the opposite of what you do when climbing. For youths and adults, this part should last for around 10 minutes; it should last for a bit longer with kids – around 20 minutes.

The specific part should be climbing based, and the most common way to start is with easy routes or boulder problems on larger holds, before gradually making the climbing more difficult. In this part of the warm-up, you can – and should – introduce the technical elements you wish to work on.

To adequately warm up your fingers with regards to injury prevention it is recommended that you do more than 100 moves before significantly increasing the difficulty. Your warm-up should also be specifically targeted towards the contents of the main part of your training session. If steep bouldering is on the agenda, the last part of the specific warm-up should involve steep climbing and you should make sure that your fingers, arms, upper body and core are warm. If you are doing vertical lead climbing, your warm-up should be characterised by warming up the fingers and incorporating some hip mobility training. In addition to preparing you physically, warming up should also prepare you mentally to perform. Because the specific part of the warm-up is like what you are going to do in the main part, this will have a positive effect on how prepared you are mentally for what will come next.

WARM-UP TIP

Begin by traversing on good holds or by doing some easy top-roping. Then move on to making the movements more dynamic. A simple exercise for this is to do double dynos between good holds while keeping your feet on the wall. Remember, climbing is about having fun and freedom of movement, so try to let loose mentally, be dynamic and play with dynos to get more flow in your movements. Our experience is that many climbers move better during the main part of their session if they take this approach during their warm-up.



MAIN PART

As the name implies, it is in this part that you conduct the training which you have planned. If you have planned to do bouldering pyramids, you should decide in advance which problems to try. If you have planned to do circuits, you can use the warm-up to test moves and sequences of different sections of the circuits, so that every attempt in the main part is as good as it can be. With technical sessions, you should predefine focus areas and what technical challenges each boulder problem or route should offer. If you want to work on weight transfer and balance, choose less-steep wall angles and holds that will not allow you to pull hard. You must either have set the problems in advance or have an overview of the various problems and routes at your gym – and you must be able to stick to this plan.

Whatever you have planned to do in the main part, you should have prepared yourself mentally for what is going to happen. There is a big difference in attitude if you're going for maximum effort on individual routes or boulder problems, versus having an endurance session on routes or a technical bouldering session. For physically demanding sessions, your stress level must be higher with a goal of maximising each attempt. For technical sessions, you need to be mentally prepared for problem-solving and practising elements you are less skilled at, while for endurance sessions you must be prepared to endure the sensation of climbing with pumped forearms.

If you are going to train multiple elements in the same main part, it is important to plan in which order to train them. The explosive, physically demanding and most technically challenging parts of the session should follow immediately after the warm-up to ensure the best possible quality of execution. For example, you can start with campusing or deadhangs after the warm-up, then go on to climbing routes or boulders, and then finish with general training and injury prevention.



FINISHING

After a hard climbing session, it can be a good idea to 'cool down' on easier routes or boulder problems to finish the session, both to add some more training into your session and to shorten the recovery time. Or you can round off your training by spending 20 minutes doing some stretching or other injury-prevention training. Although many skip the final part of the session due to a lack of time, there is often a lot to be gained here, since you can work on general strength training, mobility or technical elements that are less physically demanding.

SESSION PLANS

Below are three examples of session plans for training different skillsets. With each session plan, we have added comments related to the specific contents. By planning the sessions in this way it's easier to see the relationship between the warm-up, the main part and the finish, as well as time allocated for each part.



SESSION PLAN FOR BOULDERING: MAXIMUM STRENGTH AND FINGER STRENGTH

	WHAT	TIME
Warm-up	General part: injury-prevention exercises.	10 minutes
Specific part: deadhangs without added weight on good holds. Dynamic moves, good holds, easy bouldering.	20 minutes	Visualise individual 1
Main part	Max deadhangs.	20 minutes
Max bouldering.	30–45 minutes	Visualisation before 4
Finish	Core training.	20 minutes

This is a physically demanding workout, and this needs to be reflected in the warm-up which prepares you for the main part. The higher the intensity of the main part, the higher the intensity needs to be towards the end of the warm-up. It is especially important to warm up the fingers properly for this session, as it starts with deadhangs. Because you are combining deadhangs and max bouldering in the same session, it is important that the main part doesn't last too long, as both of these methods have a high intensity rating. Settle for four boulder problems and two attempts per problem, but make sure to maximise the quality of each attempt. The session can finish with core training, as the main part focuses mainly on the fingers, arms and upper body.

SESSION PLAN FOR ENDURANCE

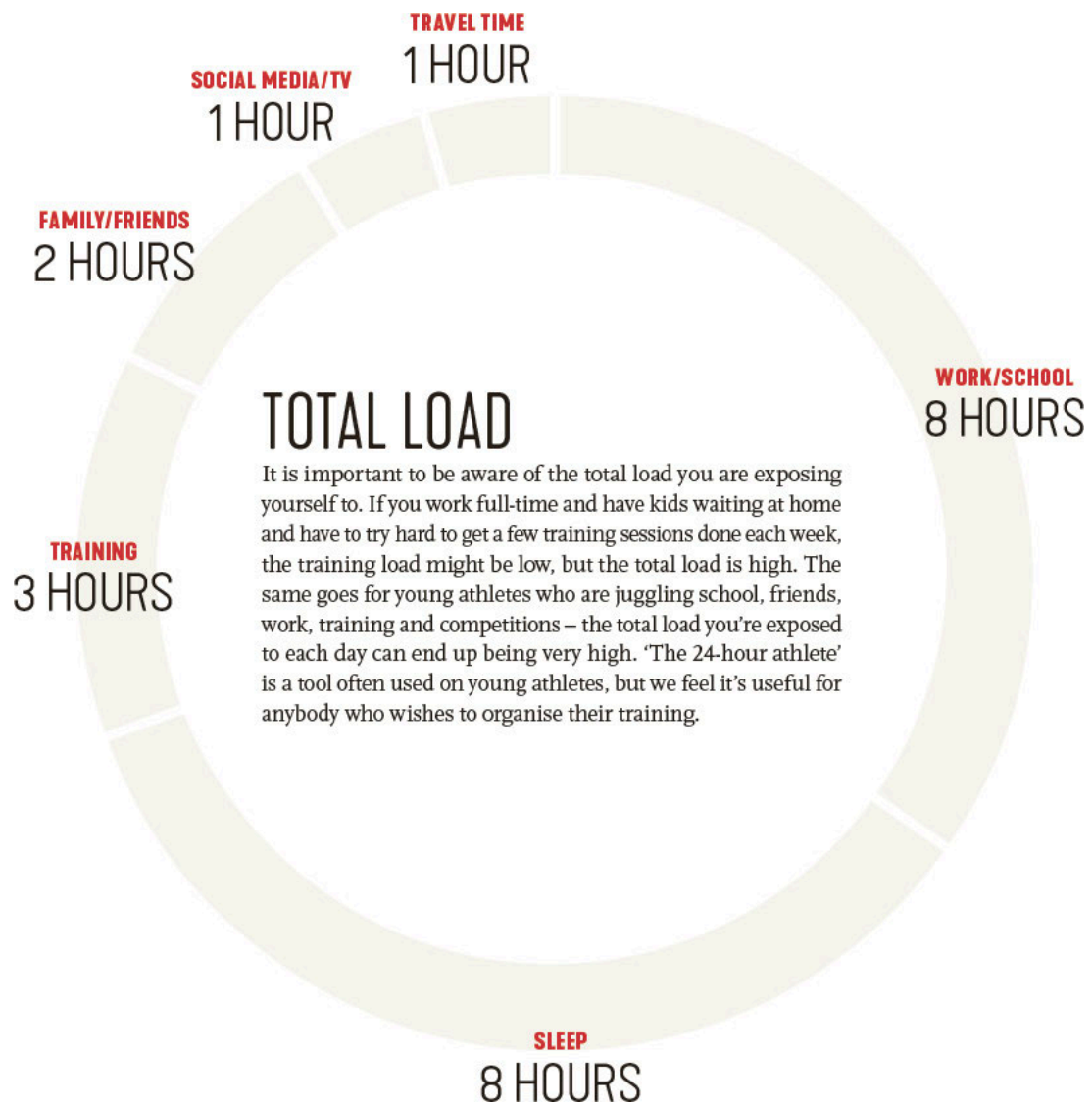
	WHAT	TIME
Warm-up	General part: skipping rope/light jog and injury-prevention exercises	15 minutes
Specific part: easy top-rope climbing (2 routes), easy lead climbing (1–2 routes).	16 minutes	Awareness of clipping
Main part	3 x 5.	90 minutes
Finish.	Cooling down and mobility training.	20 minutes

The main focus of this session is endurance training, and the main part is relatively long with low intensity. This requires less time for warming up, and the transition into the main part can be quite fluid. With so many routes in one session, you have the opportunity to focus on technical and tactical elements like weight transfer, energy-efficient climbing and clipping technique. It is important to be conscious of the feeling of getting pumped so that you can control the intensity. This will also make you more confident in climbing with an increasing level of pump in your forearms. You can finish with some easier routes to shorten your recovery time, and mobility training for your hips and thoracic spine is a good match for this session as your body is already warm and your arms are tired.

SESSION PLAN FOR TECHNIQUE: FOCUSING ON BALANCE AND WEIGHT TRANSFER

	WHAT
Warm-up	General part: skipping rope/light jog. Active mobility training.
Specific part: easy, vertical traversing. Easy, vertical boulder problems.	10 minutes
Main part	Bouldering. 10 different boulder problems requiring fronting, sidepulls, and flagging. Slabs, and vertical
Finish	Arm and upper body strength. Injury-prevention exercises.

For the general part of the warm-up, you can do active mobility training since the techniques of the main part require mobility. You can seamlessly transition from the easy bouldering in the specific part to the main part, but you should decide in advance which problems you are going to work on in the main part. A technical bouldering session requires patience and problem-solving, and none of the problems should be too physically demanding. Since the physical load has been low for the main part, you can finish the session with both specific and general strength training.





Having kids has a big impact on the total load in your life, and is definitely a variable that needs to be considered when planning for training.

TO ILLUSTRATE WHAT IT MEANS TO HAVE A CONSCIOUS AWARENESS OF THE TOTAL LOAD WE HAVE CHOSEN THE FOLLOWING THREE EXAMPLES:



STIAN CHRISTOPHERSEN FULL-TIME JOB AND FATHER OF TWO

Transitioning from prioritising climbing over everything to prioritising the kids was challenging and required me to totally transform the way I structured my training. First and foremost I had less time per training session, and the sessions had to be coordinated with work and family commitments. During periods of less sleep and greater total load, I had to prioritise what I wanted to work on in my training so that I could get the most out of each session. It was realistic to do two climbing sessions per week, supplemented by two sessions at home on the fingerboard. I decided to train finger strength and arm strength at home and vary the climbing between bouldering and lead, depending on what I wanted to train for. With the fingerboard at home, I got to train exactly what I needed in a relatively short space of time, and while at times it was hard to motivate myself for those sessions, they did produce good results. I also got more out of the two climbing sessions per week when I had decided in advance whether it should be a hard bouldering session, a high volume lead session or a more free session where the most important thing was being social and taking a break from everyday life.

Some of my hardest ascents to date I did after having children and dealing with an ever-increasing workload. The fact that I have managed to progress in times of high total load has given me confidence that I can continue to improve, and at the very least maintain my level, by being more structured in my training and prioritising more clearly what I want to work on every session.

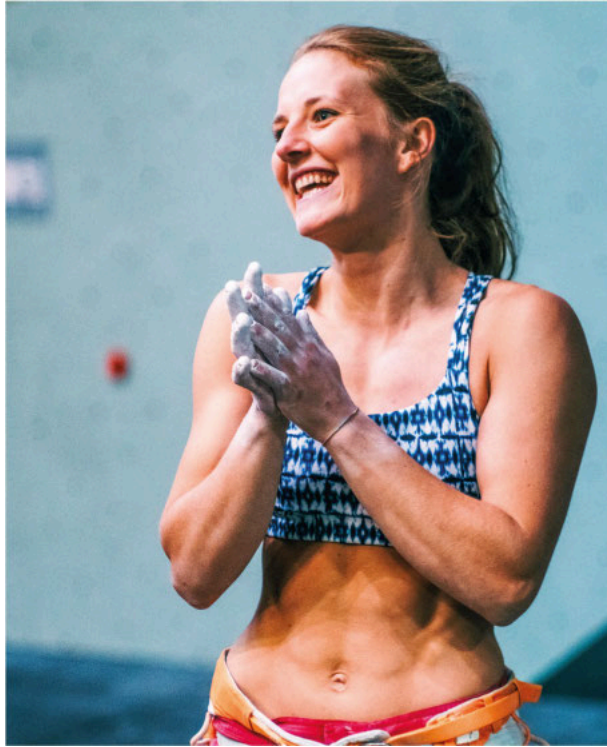


PHOTO: BJRØNAR SMESTEAD

TINA JOHNSEN HAFSAAS PROFESSIONAL CLIMBER

The biggest advantage of being a full-time climber is that I can show up for training feeling refreshed, motivated and ready to pull hard. When I was in high school and competed internationally as both a junior and a senior and had over 150 travel days, it was difficult to have the energy surplus required to train as hard as I needed in order to progress. Today I have a life where training is the priority, and everything else is planned around that.

To continue up the progression ladder I depend on having sessions of the highest quality possible, and the main risk of being a full-time climber is exercising too much. In my daily routine with 15 to 30 hours of training a week, it is important that I structure the week so that the sessions fit together, that they encompass everything I should go through, and that they're not just high-intensity sessions. It goes without saying that periods with a high training volume will include sessions of reduced quality, but this is because I have deliberately planned for a lot of training in that period. During these periods, it's important to ensure optimal recovery and that I don't lose energy as a result of a hectic daily life.

After many years of focusing on basic training, I have a body that can withstand high volume and heavy loads, but it's still important that the training doesn't become one-sided with too much attention directed to just a few elements. As long as I manage to vary the content of my training weeks and have a plan for how much I need to rest and how hard I should train during different periods of the year, it's not a problem in my everyday life that I exercise too much. I am privileged to have all the time in the world available for training.



MARIA STANGELAND FULL-TIME JOB, MARRIED, NO KIDS

I started climbing relatively late and immediately fell for the sport. It wasn't long before I was climbing three to five days a week, for two to three hours each time. Before, when I did other forms of training like running or hitting the gym, I usually had time for three hours a week. This is because I would spend half a day just trying to drag myself out of the door. With climbing, it's different. I'm literally running out of the door. I'm in my home office, already wearing my climbing clothes, and the moment I slam the MacBook shut I grab my bag and off I go. I have also become less worried about appearances and carry my bag with me everywhere, so I'm ready to go climbing directly from any situation. In meetings with strangers, it's often a nice icebreaker.

It helps to have three regular days with a regular partner or a regular crew. I then plan other things around this. I only change the regular days if I have a very good reason. They are also distributed in such a way that there are days off, so I can either recover, do alternative training, or squeeze in an extra climbing session, depending on what's suitable and what the total load is.

Climbing is also a very social sport, with nice people who share my passion, and so it's also become a haven where I enjoy spending time with good friends and acquaintances. I climb because I want to, not because I absolutely need to exercise. Besides, I usually plan my sessions as early as possible in the afternoon so I can go straight from work and have the opportunity to do something else afterwards. Unfortunately, early morning training is not for me, so instead I work early.

It doesn't cost me anything to squeeze in some climbing at times when I otherwise wouldn't think to train. I surround myself with people who are just as obsessed with climbing as I am, who, in addition to sticking to the same invariable plans, are also able to join a session at the spur of the moment. There are more of them than you'd think. And I tell my pack why it's so important for me to prioritise climbing. For me, it's necessary stress therapy, meditation and mental training, and it gives me an invaluable sense of achievement. It's incredible how people who care about you can be flexible and solution-oriented if they understand why you're doing something.



HOW TO MAINTAIN YOUR LEVEL WHEN THINGS ARE HECTIC

Maintaining your level requires a lot less effort than trying to raise it. Raising your level requires a lot of training over a long period of time, and at times this is not practically possible. You can still maintain the level you are at by avoiding complete breaks from climbing and loading the fingers, arms and upper body during these periods. The most important thing is to not go completely without exercise for more than two weeks. It is then that the training base you have built up will start to gradually break down and you will need to start training at a lower level to reduce the risk of injury. A simple aid is a fingerboard at home or on a trip that allows you to hang by your fingers, do pull-ups and train your abdominal and back muscles in hanging positions. Completing just a few short workouts a week makes it possible to exercise less, yet to a satisfactory level, during hectic periods. You will maintain continuity in your training, thus reducing the risk of injury and ensuring that you are well prepared for periods when you have time for more and harder training.

SUMMARY

Planning for training lets you structure your day-to-day training, define which factors to prioritise, create a balance between loading and recovery, and stimulate peaking. All in all, this will provide better quality and continuity in your training and allow you to train and climb more with a lower risk of chronic injuries.



Cecilie Skog climbing during her last session before giving birth to her daughter Vega three days later.

A CLIMBER'S STORY

Norwegian climber Cecilie Skog (b. 1974) is best known for her expeditions to the poles and the Himalaya, but she's also an avid sport climber. She's a registered nurse and mother of two girls, Vilja and Vega.

CLIMBING DURING AND AFTER PREGNANCY

‘Cecilie, can you write a chapter in the book about climbing while pregnant?’

My basic positive and spontaneous response to such questions is usually YES! Since the request this time also came from good friends, answering no was unthinkable. Besides, I have almost two years of experience of climbing while pregnant.

When I sat down to give my advice and recommendations, I have to admit I found it difficult. After all, I only have my own very subjective experiences.

How could I provide valuable tips without knowing who you are, what kind of pregnancy you are going through and how much you have climbed before? Nor am I a midwife or an obstetrician. I'm just a nurse.

It is amazing how vulnerable one suddenly feels when a tiny little miracle is growing inside your belly. We only want the best for the child who is trapped for nine months in somebody else's body.

The female body is genetically coded to change and adapt when pregnant. Our sense of taste and smell changes. We get more tired and tired more often. We undergo hormonal changes, blood volume increases, bloodflow increases, and ligaments become looser and softer. Our assessment of consequences also changes. We adapt to the situation with plenty of help from nature.

And the health clinic. They provide us with answers to the thousands of questions that suddenly arise. Whether they can provide in-depth answers to your questions about pregnancy and climbing is unknown. I never asked them such questions: I went to the source – my female climbing friends. Those who had climbed through one pregnancy after another. They had the answers. There were many I could ask since I was last of the bunch. Since mine was a geriatric pregnancy, because of my old age, I might as well have been a grandmother to the child I was carrying.

As I write this now, I have decided that I'm not going to preach one way or the other. I don't want to systematically set up a training programme for pregnant women or provide a definite and final answer.

But I can share my story.

During my two pregnancies, I climbed two to three days a week throughout each one. Vilja arrived a few weeks early, in week 37. The last session we 'hung out' together was the day before she came into this world. I sold my pregnancy climbing harness online just a couple of days before baby sister Vega arrived right on schedule, on my birthday, in week 36.

Climbing while pregnant meant that my patient climbing partners and I endured many long sessions. Because of all the pee breaks. As my belly grew, the climbing was interrupted more and more frequently by these necessary breaks. I could hold on for hours. All these interruptions gave me stamina and a sense of achievement that was good to carry with me during the day. For the most part, I climbed inside. I would often sneak into Vulkan, a climbing wall in Oslo, before the capital woke up and before the actual opening hours of the gym.

‘THE BELLY GREW GRADUALLY. THE KILOS CAME GRADUALLY. GRADUALLY I ALSO DEVELOPED A NEW WAY OF CLIMBING.’

Actually, I like people a lot and am used to sticking my nose out and asking to be seen here and there. So I was surprised to find that I suddenly didn't like the attention as much.

Normally, without a baby in my belly, I have the body of a 14-year-old boy. No matter how exquisitely I curled my hair, all of a sudden it was my breasts and belly that entered the room first, that got the attention, the looks. A while later I would arrive. Somewhat embarrassed to learn that eye contact wouldn't be an option until several seconds later. I've always wanted adequately large breasts – who hasn't? But, when they did first grow out, I thought that these huge knockers looked really out of place on my body.

Commenting on pregnant bodies is widely accepted. This was a new discovery. Apparently, it's OK to have opinions on them, loudly and in front of others.

'Yes, now I can see you're pregnant!'

'Oh my, how big you have become!'

'Ready to burst at the seams!'

'Wow, you're climbing? With that belly?'

Therefore it was a privilege to climb in peace in the mornings, even though I don't really suffer from social anxiety.

I felt comfortable climbing on lead until about week 16, after which I only climbed on top rope. It would often be my breathing that was the limiting factor, not the usual pump in my forearms. I stopped bouldering quite early in pregnancy because I felt uncomfortable with falling or dropping on to the crash pads.

The belly grew gradually. The kilos came gradually. Gradually I also developed a new way of climbing. The adaptations of my climbing technique with my new body wasn't something I was focusing on consciously. It just happened. Naturally. I moved more sideways. Twisted in more to use my abdominal muscles. I had to place more emphasis on weight transfer. Literally. I had to compensate for my lack of ability placing my feet up high. I had to do more and shorter moves. I had to position myself differently.

Incredibly enough I felt good when climbing. On the ground I was stomping around like a cow – on the wall I felt completely different. It felt wonderful to stretch my back, use my arms, lessen the load on my feet. Stand on my toes. Sometimes I felt like I was dancing. Maybe it was the contrast with waddling around on the ground that made me feel like I was weightless on the wall. Sometimes I just had to exclaim my enthusiasm with a 'weee, I'm a helium balloon!'

Little joys can also be hard to hide.

After giving birth, both of them by caesarean, I waited six weeks before starting to climb again. Vilja was most excited about moving around. All the time. While on maternity leave with her we did a hundred thousand kilometres of strolling and not much climbing. Vega was apparently happy to lie on a pad marinated in chalk along with eight to ten other toddlers. Fortunately. Without hair they lay there like a nice, homogenous bunch without any motor skills, looking at the nice, colourful climbing holds. We bragged about them and said they were patient. We formed an incredible women's community on and underneath the walls. We cheered each other on. Shared beta. We breastfed. We cuddled the little ones. Our own and each other's. If somebody cried, comfort crisscrossed the group. And we drank coffee. And we gossiped. And breastfed some more. Good times. We were maternity bouldering two, three times a week. For the first few months, I paid attention to my operation wound with scar tissue and a tender abdomen. Surprisingly, although the same thing happened after the first pregnancy, I wasn't stronger in my comeback, even though I was several kilos lighter.

'AFTER A FULL YEAR OF MATERNITY LEAVE AND MATERNITY BOULDERING, I WAS STRONGER THAN EVER.'

It's possible expectations were high after all the comments about how I was going to fly up the wall after the baby was out. People I met at the climbing walls probably thought it had to be incredibly hard to vertically move a body the size of a small hippopotamus. People tilted their heads slightly as a sign of sympathy and wanted to tell me that all the exercise I put in would pay dividends. Tandem climbing was the same as using a weight vest. Slowly but surely, I adopted the expectation that gravity would cease as my girl was out.

The brutal truth was that I felt more at home on the wall with my daughter as a passenger. I felt stronger during the last two months of pregnancy than the first two as a mother. Even though I was heavier with a child, a placenta, and all the amniotic fluid, at least I had the feeling of filling my own body. Tight and firm. After giving birth I wasn't feeling very well put together. I wasn't in touch with my abdominal muscles and looked like a well-used bean bag.

Slowly but surely we got stronger. All the mothers. And the babies on the pads. We loved the feeling of mastery. Big and small. After a full year of maternity leave and maternity bouldering, I was stronger than ever. Maybe for the first time in my life I had real continuity in my climbing. In the 20 years I've been climbing, my activity level has always been a bit on and off. Something I've done between trips. Every time after a trip, I felt that I was starting all over again. Now the climbing was organised, with appointments three times a week. A motivating development. Extra motivational when it happens together.

Pregnancy climbing and maternity bouldering is a field that hasn't seen much research. I know nothing about consequences, injuries and long-term effects. As far as I know, nobody has written any papers on the subject.

If you are wondering whether to ask an open question about climbing during pregnancy in one of the many women's forums online – don't. You'll either be decapitated or crucified with arguments about how bad you will be as a mum. How selfish you are to consider doing extreme sports while pregnant, and that you should now think of others and not yourself. That last one is actually correct. But, regular physical activity improves blood circulation. By climbing you increase muscle mass which provides more oxygen throughout the body. Increased bloodflow to the mother during exercise also provides increased bloodflow to the foetus. Physical activity reduces

the risk of weight gain and gestational diabetes and is probably important in preventing pre-eclampsia (pregnancy poisoning). Also, when we exercise, we secrete tons of endorphin hormones. The body's own morphine, the happiness hormone. So, for several hours after a session on the wall, we can walk around with a feeling of ecstasy. All this euphoria then trickles down the chain of command. In this case, to the baby in the belly.

Because I have had such pleasure from climbing as a pre- and post-pregnancy activity, it's hard to not want to recommend it to anybody who asks.

GOOD LUCK!

Note from the authors: We want to emphasise that neither of us are doctors, gynaecologists or midwives, or have any personal experience with being pregnant – obviously. There isn't much data on climbing and pregnancy, and we have therefore chosen not to present any absolute advice or opinions on what you should – or should not – do. Our experiences align with Cecilie's story, and with reasonable consideration regarding falling when bouldering or lead climbing, it seems most people can continue climbing for as long as they are comfortable doing so. If you have any specific questions or are unsure if there are any special precautions you need to take during your own pregnancy, please contact a doctor, midwife or your health clinic.
